

Certificate_Validator

Comprehensive Project File Architecture, Purpose & Utility Catalog

Documentation Overview: This document provides an exhaustive, line-item reference of **every single file and directory** across the Certificate_Validator codebase (148 files total). For each file, this report details its precise role, architectural purpose, and functionality within the offline-first academic credential verification platform.

1. Root Workspace Configuration & Documentation

File Path & Name	Functionality & Purpose in Certificate_Validator
<code>.gitignore</code>	Specifies system files, node_modules, build outputs, and database runtime files to be ignored by Git source control.
<code>package.json</code>	Master npm workspace manifest defining project scripts (dev, build, test) and workspace setup across backend and frontend.
<code>package-lock.json</code>	Automatically generated lockfile pinning exact dependency versions to ensure reproducible installations across environments.
<code>README.md</code>	Master repository documentation featuring quick start instructions, problem statement, cryptographic architecture, user roles, and UI screenshots.
<code>Certificate_Validator_Project_Structure.pdf</code>	Exported executive PDF architectural report summarizing directory hierarchy and cleanup results.

2. Documentation Directory (docs/)

File Path & Name	Functionality & Purpose in Certificate_Validator
docs/screenshots/certificate-template.jpg	Documentation visual preview graphic representing issued academic certificate design.
docs/screenshots/fraud-analysis-modal.png	Documentation visual screenshot showcasing the interactive 8-point AI fraud risk analysis modal.
docs/screenshots/platform-overview.png	Documentation screenshot of the CredentialVault landing page and feature highlights.
docs/screenshots/verified-status-stamp.png	Documentation screenshot demonstrating the verification result interface with VALID stamp overlay.
docs/templates/01_graduation.jpg	High-resolution background template graphic used for degree graduation certificates.
docs/templates/02_distinction.jpg	Background template graphic used for graduation with distinction honors awards.
docs/templates/03_merit.jpg	Background template graphic used for academic merit recognition certificates.
docs/templates/04_academic_excellence.jpg	Background template graphic used for Dean's list and academic excellence awards.
docs/templates/05_course_completion.jpg	Background template graphic used for short-course and training completion certificates.
docs/templates/06_internship.jpg	Background template graphic used for corporate and institutional internship completion certificates.
docs/templates/07_project.jpg	Background template graphic used for capstone and major project completion credentials.
docs/templates/08_bonafide.jpg	Background template graphic used for institutional bonafide student status certificates.
docs/templates/09_participation.jpg	Background template graphic used for conference, seminar, and workshop participation certificates.

3. Backend Environment & Core Server (backend/)

File Path & Name	Functionality & Purpose in Certificate_Validator
backend/.env	Local environment file defining server port, secret JWT keys, database paths, and AI provider configurations.
backend/.env.example	Template environment file documenting required environment variables for deployment.
backend/.gitignore	Backend-specific gitignore excluding node_modules, local SQLite databases, uploads, and secret environment files.
backend/database.sqlite	Production SQLite database file storing persistent tables (users, certificates, revocations, audit logs, learning goals).
backend/database.sqlite-shm	SQLite Write-Ahead Logging (WAL) shared memory index file for concurrent database reads/writes.
backend/database.sqlite-wal	SQLite Write-Ahead Log journal file ensuring atomic database transactions and crash resilience.
backend/package.json	Backend package manifest listing Node.js dependencies (Express, better-sqlite3, jsonwebtoken, bcrypt, uuid, pdfkit, qrcode).
backend/package-lock.json	Backend locked dependency tree file ensuring reproducible server deployments.
backend/server.js	Main Express server application entry point configuring middleware, registering all 15 API route modules, and binding port 5000.
backend/config/db.js	Database connection initializer, table schema definition script, and index creation logic for SQLite database.

4. Backend API Controllers (backend/controllers/)

File Path & Name	Functionality & Purpose in Certificate_Validator
backend/controllers/analyticsController.js	Handles API requests for institutional issuance metrics, department breakdowns, and verification scan activity trends.
backend/controllers/auditController.js	Handles compliance audit log retrieval, keyword filtering, action category searches, and CSV log exports.
backend/controllers/authController.js	Handles user registration (University & Student), credential login, password hashing via bcrypt, and JWT authentication tokens.
backend/controllers/blockchainController.js	Handles queries for simulated EVM blockchain ledger blocks, transaction hashes, and block explorer data.
backend/controllers/certificateController.js	Handles certificate creation, bulk Excel processing, PDF document generation, and QR code embedding.
backend/controllers/chatAssistantController.js	Processes natural language queries sent to the AI assistant and formats contextual answers.
backend/controllers/disclosureController.js	Handles creation and public verification of RSA-2048 signed selective claim disclosures (e.g. CGPA >= 3.5).
backend/controllers/fraudDetectionController.js	Triggers and returns 8-point AI fraud detection risk scores, OCR consistency checks, and anomaly reports.
backend/controllers/goalController.js	Manages student learning goals, priority settings, target dates, and 1-check-in-per-day habit streak tracking.
backend/controllers/passportController.js	Manages creation, updates, and public retrieval of student Digital Skill Passport profiles.
backend/controllers/revocationController.js	Executes cryptographic certificate revocations, RSA signature generation over revocation records, and status lookups.
backend/controllers/templateController.js	Returns certificate SVG/HTML visual layout templates and category metadata.
backend/controllers/universityController.js	Manages university registrar profiles and public RSA key distribution endpoints for verifiers.
backend/controllers/verificationController.js	Executes multi-step verification pipeline (RSA-2048 signature, SHA-256 hash integrity, replay protection, revocation status).
backend/controllers/walletController.js	Returns student credential wallet data, issued certificate lists, signature verification statuses, and activity summaries.

5. Backend Middleware, Models, Routes & Services

File Path & Name	Functionality & Purpose in Certificate_Validator
backend/middleware/auth.js	JWT authentication middleware (`authenticateToken`) verifying bearer tokens and enforcing role-based access control.
backend/middleware/rateLimiter.js	API rate-limiting middleware restricting request bursts to protect verification endpoints against DDoS attacks.
backend/middleware/upload.js	Multer upload middleware configuring temporary storage for Excel bulk certificate files.
backend/models/chatHistoryModel.js	Data model for persisting user conversation sessions with the AI chat assistant.
backend/models/fraudAnalysisModel.js	Data model for saving and loading AI fraud evaluation results, anomaly flags, and confidence scores.
backend/models/passportModel.js	Data model for structuring and retrieving student skill passport records (bio, skills, projects, licenses).
backend/models/templateModel.js	Data model defining visual layout attributes and text coordinates for certificate category templates.
backend/routes/analyticsRoutes.js	Express router mounting analytics endpoints under `/api/analytics`.
backend/routes/auditRoutes.js	Express router mounting compliance audit log endpoints under `/api/audit`.
backend/routes/authRoutes.js	Express router mounting authentication endpoints (`/login`, `/register`) under `/api`.
backend/routes/blockchainRoutes.js	Express router mounting blockchain ledger endpoints under `/api/blockchain`.
backend/routes/certificateRoutes.js	Express router mounting certificate issuance and PDF download endpoints under `/api/certificates`.
backend/routes/chatAssistantRoutes.js	Express router mounting AI assistant endpoints under `/api/chat`.
backend/routes/disclosureRoutes.js	Express router mounting selective claim disclosure endpoints under `/api/disclosure`.
backend/routes/fraudRoutes.js	Express router mounting AI fraud detection analysis endpoints under `/api/fraud`.
backend/routes/goalRoutes.js	Express router mounting student learning goal and habit tracker endpoints under `/api/goals`.
backend/routes/passportRoutes.js	Express router mounting Digital Skill Passport endpoints under `/api/passport`.
backend/routes/revocationRoutes.js	Express router mounting certificate revocation endpoints under `/api/revocation`.
backend/routes/templateRoutes.js	Express router mounting certificate template endpoints under `/api/templates`.
backend/routes/universityRoutes.js	Express router mounting university registrar profile endpoints under `/api/university`.
backend/routes/verificationRoutes.js	Express router mounting certificate verification endpoints under `/api/verify`.
backend/routes/walletRoutes.js	Express router mounting student wallet endpoints under `/api/wallet`.
backend/services/aiRiskScoringService.js	Evaluates certificate data against an 8-point security check rule matrix to generate a normalized fraud risk score (0-100).
backend/services/chatAssistantService.js	Processes incoming user messages and generates domain-aware answers about verification, cryptography, and credentials.
backend/services/contextBuilder.js	Constructs structured system context prompts containing platform metadata for LLM processing.

backend/services/fraudDetectionService.js	Core engine coordinating OCR text extraction, layout matching, and risk evaluation for uploaded certificate files.
backend/services/llmProvider.js	Multi-provider LLM abstraction layer supporting OpenAI GPT-4, Google Gemini, Anthropic Claude, Azure, and heuristic fallback.
backend/services/ocrService.js	Tesseract OCR service extracting text strings from scanned certificate images for automated verification.
backend/services/passportService.js	Business logic service structuring student skills, verified credentials, and portfolio links into a shareable passport.
backend/services/promptBuilder.js	Helper module building specialized prompts for AI risk analysis and verification assistance.
backend/services/templateService.js	Dynamic rendering engine injecting student data into HTML/SVG certificate layouts for PDF generation.
backend/tests/crypto.test.js	Automated unit test suite verifying RSA-2048 key generation, SHA-256 hashing, and digital signature validation.
backend/tests/disclosure.test.js	Automated unit test suite verifying creation, RSA signing, and verification of selective predicate disclosures.
backend/tests/revocation.test.js	Automated unit test suite verifying cryptographic revocation signing and revocation status checks.
backend/tests/verification.test.js	Automated unit test suite verifying end-to-end verification pipeline, replay protection, and tamper detection.
backend/uploads/.gitkeep	Placeholder file ensuring the uploads directory is tracked by Git while uploaded QR image files are ignored.
backend/utils/auditLogger.js	Utility module recording tamper-evident audit log entries into the database for all key platform actions.
backend/utils/blockchain.js	Simulated EVM blockchain ledger engine managing block generation, transaction hashing, and immutability checks.
backend/utils/crypto.js	Core cryptographic helper module wrapping Node.js `crypto` for RSA-2048 key pair generation, SHA-256 hashing, and signing.

6. Frontend Core & Assets (frontend/)

File Path & Name	Functionality & Purpose in Certificate_Validator
frontend/.gitignore	Frontend gitignore excluding build output (`dist/`), node_modules, and environment files.
frontend/eslint.config.js	ESLint configuration file enforcing code style standards and React hook rules.
frontend/index.html	Main HTML template containing root DOM element (`#root`), viewport configuration, and web app title.
frontend/package.json	Frontend dependencies manifest (React 19, Vite, Framer Motion, Recharts, jsPDF, html2canvas, lucide-react).
frontend/package-lock.json	Locked frontend dependency tree file ensuring identical client package builds.
frontend/README.md	Frontend-specific developer setup guide and usage documentation.
frontend/vite.config.js	Vite build tool configuration setup with React plugin, PWA service worker settings, and API proxy routes.
frontend/public/favicon.svg	SVG browser favicon icon.
frontend/public/icons.svg	Vector icon sprite sheet used throughout the frontend user interface.
frontend/public/pwa-icon-192.jpg	192x192 PWA application icon for mobile home screen installations.
frontend/public/pwa-icon-512.jpg	512x512 PWA application icon for high-DPI device splash screens.
frontend/src/App.jsx	Root React application component managing client-side routing (`BrowserRouter`), loading contexts, and top-level modals.
frontend/src/index.css	Primary CSS stylesheet containing design system CSS tokens, typography, glassmorphism utilities, dark mode, and animations.
frontend/src/main.jsx	React DOM render entry point mounting the App component into the HTML root element.
frontend/src/styles/wallet.css	Custom CSS styles specific to the student credential wallet, 3D card perspective, and habit tracker cards.

7. Frontend API, Hooks & Utilities

File Path & Name	Functionality & Purpose in Certificate_Validator
frontend/src/api/analytics.js	API integration functions for fetching university analytics and verification volume stats.
frontend/src/api/audit.js	API integration functions for fetching and filtering institutional audit log entries.
frontend/src/api/blockchain.js	API integration functions for querying blockchain blocks and transaction ledger details.
frontend/src/api/chat.js	API integration functions for communicating with the AI chat assistant endpoint.
frontend/src/api/client.js	Base Axios HTTP client with automated authorization header insertion and error interceptors.
frontend/src/api/disclosure.js	API integration functions for requesting and verifying selective disclosure proof links.
frontend/src/api/fraud.js	API integration functions for triggering AI fraud risk scoring on uploaded certificate files.
frontend/src/api/passport.js	API integration functions for updating and retrieving student skill passport data.
frontend/src/api/templateApi.js	API integration functions for fetching certificate visual templates.
frontend/src/api/wallet.js	API integration functions for fetching student wallet statistics, certificates, and portfolio links.
frontend/src/hooks/useAIProvider.js	Custom React hook for checking active AI provider status (Gemini, OpenAI, Heuristic) and managing settings.
frontend/src/hooks/useHeaderHeight.js	Custom React hook dynamically computing top header heights for responsive page layouts.
frontend/src/hooks/useReveal.js	Custom React hook using IntersectionObserver to trigger smooth Framer Motion entrance animations on scroll.
frontend/src/utils/certificateCategory.js	Configuration module mapping certificate categories to color schemes, icons, and template IDs.
frontend/src/utils/certificatePdf.js	Utility function rendering React DOM certificate nodes into downloadable PDF files via html2canvas & jsPDF.
frontend/src/utils/excelParser.js	Utility helper parsing uploaded Excel spreadsheets into JSON objects for bulk certificate issuance.
frontend/src/utils/keyCache.js	Caching module storing issuer RSA public keys in LocalStorage/IndexedDB for offline verification.
frontend/src/utils/offlineCrypto.js	Browser Web Crypto API wrapper verifying RSA signatures and SHA-256 hashes completely offline.
frontend/src/utils/qrcode.js	Utility component scanning and decoding QR code image files into certificate JSON payloads.
frontend/src/utils/revocationCache.js	Local caching module storing revocation lists on client device for offline revocation checks.
frontend/src/utils/walletStore.js	State persistence helper saving student credential wallet filtering and layout preferences.

8. Frontend Application Pages (frontend/src/pages/)

File Path & Name	Functionality & Purpose in Certificate_Validator
frontend/src/pages/AuditLog.jsx	Page component rendering searchable compliance audit log table with filter tabs and CSV export.
frontend/src/pages/BlockchainExplorer.jsx	Page component visualizing simulated EVM blockchain blocks, transaction lists, and immutability status.
frontend/src/pages/DigitalSkillPassport.jsx	Page component allowing students to edit bio, manage technical skills, showcase projects, and view verified badges.
frontend/src/pages/Home.jsx	Landing page featuring platform overview, cryptographic architecture highlights, and role quick-access links.
frontend/src/pages/PublicDisclosureView.jsx	Public verification page allowing anyone to independently verify an RSA-signed predicate disclosure claim.
frontend/src/pages/PublicSkillPassport.jsx	Publicly shareable view of a student's verified digital skill passport portfolio.
frontend/src/pages/Register.jsx	Portal choice page allowing users to select University or Student registration flow.
frontend/src/pages/StudentAnalytics.jsx	Student analytics dashboard displaying credential achievements, skill progress, and activity breakdown.
frontend/src/pages/StudentDashboard.jsx	Student dashboard displaying list of earned certificates, request options, and skill passport quick links.
frontend/src/pages/StudentLogin.jsx	Student authentication login page with email and password fields.
frontend/src/pages/StudentRegistrar.jsx	Student account registration page with university selection linkage.
frontend/src/pages/TemplateManager.jsx	University template preview page allowing registrars to inspect certificate designs before issuance.
frontend/src/pages/UniversityAnalytics.jsx	University analytics dashboard featuring issuance charts, department breakdown pie charts, and revocation metrics.
frontend/src/pages/UniversityDashboard.jsx	Master university dashboard for issuing single/bulk certificates, managing student records, and revoking credentials.
frontend/src/pages/UniversityLogin.jsx	University registrar authentication login page.
frontend/src/pages/UniversityRegistrar.jsx	University registration page featuring automated RSA-2048 key pair generation on signup.
frontend/src/pages/VerificationAnalytics.jsx	Verification analytics dashboard tracking scan volumes, pass/fail ratios, and geographic verification activity.
frontend/src/pages/Verifier.jsx	Public verifier portal supporting Certificate ID lookup, QR scan, file upload, offline verification, and AI fraud checks.
frontend/src/pages/WalletDashboard.jsx	Student credential wallet page featuring 3D-tilt cards, Achievement Timeline pop-up modal, Portfolio Links, and Learning Goal Tracker.

9. Frontend UI Components (frontend/src/components/)

File Path & Name	Functionality & Purpose in Certificate_Validator
frontend/src/components/AuditLogDecorations.jsx	Background vector decorations and graphics for the Audit Log page.
frontend/src/components/AuthBackgroundDecorations.jsx	Background vector decorations for authentication pages (login & registration).
frontend/src/components/CertificateTemplate.jsx	Universal visual certificate renderer outputting formatted certificate documents with QR codes.
frontend/src/components/FraudAnalysisModal.jsx	Interactive modal presenting 8-point AI fraud check items, anomaly badges, and risk score meter.
frontend/src/components/GuidedTourModal.jsx	Interactive platform onboarding tour guiding first-time users through key features.
frontend/src/components/StudentDashboardDecorations.jsx	Background vector graphics for the Student Dashboard page.
frontend/src/components/UniversityDashboardDecorations.jsx	Background vector graphics for the University Dashboard page.
frontend/src/components/VerifierBackgroundDecorations.jsx	Background vector graphics for the Verifier portal page.
frontend/src/components/AIChat/ChatBubble.jsx	Renders individual AI chat assistant message bubbles with markdown and action tags.
frontend/src/components/AIChat/ChatWindow.jsx	Floating chat window container managing conversation history, user input, and AI responses.
frontend/src/components/AIChat/FloatingAIButton.jsx	Floating action trigger button to toggle the AI Assistant window open and closed.
frontend/src/components/AIChat/SuggestedQuestions.jsx	Clickable suggested prompt chips for quick AI assistant guidance.
frontend/src/components/AIChat/TypingIndicator.jsx	Animated dots component indicating AI assistant is actively generating an answer.
frontend/src/components/decorations/BlockchainExplorerDecorations.jsx	Background graphics tailored for the Blockchain Explorer page.
frontend/src/components/decorations/StudentAnalyticsDecorations.jsx	Background graphics tailored for the Student Analytics page.
frontend/src/components/decorations/UniversityAnalyticsDecorations.jsx	Background graphics tailored for the University Analytics page.
frontend/src/components/decorations/VerificationAnalyticsDecorations.jsx	Background graphics tailored for the Verification Analytics page.
frontend/src/components/decorations/WalletDashboardDecorations.jsx	Background graphics tailored for the Student Wallet Dashboard.
frontend/src/components/motion/CountUp.jsx	Animated number counter component smoothly incrementing statistics on scroll into view.
frontend/src/components/motion/index.js	Module re-export file simplifying imports of motion primitives.

frontend/src/components/motion/RevealOnScroll.jsx	Animation wrapper component triggering Framer Motion reveal effects when scrolled into view.
frontend/src/components/motion/SkeletonCard.jsx	Skeleton card placeholder component displayed while data loading is in progress.
frontend/src/components/templates/AcademicExcellenceTemplate.jsx	Visual certificate template layout for Academic Excellence awards.
frontend/src/components/templates/BonafideTemplate.jsx	Visual certificate template layout for Bonafide Student certificates.
frontend/src/components/templates/CategoryCertificateTemplate.jsx	Dynamic wrapper component selecting and rendering the appropriate certificate template based on category.
frontend/src/components/templates/CourseCompletionTemplate.jsx	Visual certificate template layout for Course Completion credentials.
frontend/src/components/templates/DistinctionTemplate.jsx	Visual certificate template layout for Graduation with Distinction awards.
frontend/src/components/templates/GraduationTemplate.jsx	Visual certificate template layout for Degree Graduation diplomas.
frontend/src/components/templates/InternshipTemplate.jsx	Visual certificate template layout for Internship Completion certificates.
frontend/src/components/templates/MeritTemplate.jsx	Visual certificate template layout for Merit Recognition awards.
frontend/src/components/templates/ParticipationTemplate.jsx	Visual certificate template layout for Event Participation certificates.
frontend/src/components/templates/ProjectTemplate.jsx	Visual certificate template layout for Project Completion credentials.
frontend/src/components/templates/TemplateSelector.jsx	Interactive selector component allowing registrars to preview and pick certificate templates.
frontend/src/components/wallet/AchievementTimeline.jsx	Compact achievement milestone timeline featuring expandable full-screen pop-up modal browser.
frontend/src/components/wallet/CertDetailModal.jsx	Detailed modal displaying full certificate metadata, signature validity status, QR preview, and PDF download button.
frontend/src/components/wallet/EarningGoalTracker.jsx	Goal & habit tracker component enabling goal creation, target date picking, and once-per-day habit streak check-ins.
frontend/src/components/wallet/PortfolioLinksCard.jsx	Editable card component for students to manage portfolio website links and social profiles.
frontend/src/components/wallet/WalletCertCard.jsx	Interactive 3D-tilt credential card component representing individual certificates in the student wallet.
frontend/src/components/wallet/WalletDetailPane.jsx	Detail panel component displaying selected certificate info within master-detail layout.
frontend/src/components/wallet/WalletIndexList.jsx	Index list panel component displaying searchable list of certificates within master-detail layout.